Where to Look From

Knowing Whether, a Sensing Action with a Modal Precondition, and Two Lookouts, One of Them Perhaps Screened

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A robot must know whether a zone is blocked. It can see the zone from either of two lookouts, and a screen may stand in front of one of them, never both; the robot does not know which. A look through a screen decides nothing, and a look from lookout k is written with the precondition that the robot know the view from k is clear. We prove that this domain has no conformant plan: no sequence of its actions is applicable in every world and reaches the goal, because the precondition of every look is false in two of the six worlds. A policy of five nodes reaches the goal in all six, and drives to the second lookout in exactly the two worlds with a screen at the first. In the world, any sequence either drives to the second lookout in every run or leaves the zone unobserved behind a screen. Thirty paired seeds per formulation in Gazebo measure this: the policy's success is backed by the map the robot built in 30 of 30 runs, and it drives to the second lookout in 10; the classical sequence committed to the first lookout reports success in 30 and is backed in 20, every unbacked run behind a screen with no cell of the zone observed (McNemar's exact $p = 0.002$); the sequence through both lookouts is backed in 30 and slower in every pair, by a Hodges–Lehmann estimate of 26.6 s.

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1 Scope

This report states one scenario of a harness for repeated simulator runs, and the film recorded of it for the Epistemic Robotics site. The scenario, its launch file, the harness that runs and measures it and the batch of runs are not in this repository; the domain is restated here in notation, and every figure is computed from the batch's table of runs. The film's four takes were recorded with the harness unchanged.

The question the scenario asks is narrow. A look that can end without deciding is a problem when noise is its cause: an outcome that may occur in any world leaves no policy that guarantees the goal. Here the undecided look has a cause the robot can observe, a screen across the sight line, and the planner can choose where to look from. The planning is epistemic planning over dynamic epistemic logic [2], with a goal of knowing whether [3, 6].

2 The domain

One agent, $r1$. Five atoms: *blocked-z*, the zone is blocked; *tapado-1* and *tapado-2*, a screen hides the zone from lookout 1 or 2; *at-m*$_1$ and *at-m*$_2$, the robot is at lookout 1 or 2. Five actions, each an event model in the sense of Baltag, Moss and Solecki [1]:

action	kind	precondition	events
goto-m1, goto-m2	public ontic	$\neg$*at-m*$_k$	*at-m*$_k$ becomes true and the other false
vista-1	sensing	*at-m*$_1$	e-vista1-tapada where *tapado-1*, e-vista1-despejada where $\neg$*tapado-1*
mirar-1	sensing	*at-m*$_1 \wedge K_{r1}\neg$*tapado-1*	e-mirar1-bloqueada where *blocked-z*, e-mirar1-libre where $\neg$*blocked-z*
mirar-2	sensing	*at-m*$_2 \wedge K_{r1}\neg$*tapado-2*	the same, from lookout 2

The sensing actions are semi-private: $r1$ distinguishes the events, and any other agent would see only that the action took place. The initial model is the six valuations of (*blocked-z*, *tapado-1*, *tapado-2*) with $\neg$(*tapado-1* $\wedge$ *tapado-2*) and $\neg$*at-m*$_1 \wedge \neg$*at-m*$_2$, all six designated, and $r1$'s relation the universal one; the frame is S5. The goal is Kw_{r1} *blocked-z*, that is, K_{r1}*blocked-z* $\vee K_{r1}\neg$*blocked-z*. That at most one view is screened, being true in every world of an S5 model in which $r1$ relates all worlds, is common knowledge.

There is no action that checks the view from lookout 2. The two propositions below show why none is needed.

3 The hall

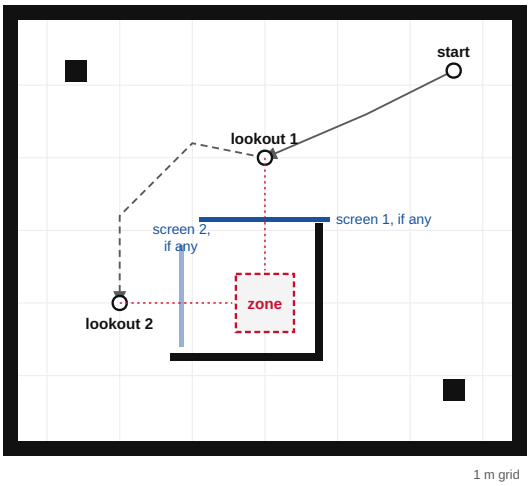


Figure 1: The hall, 7 by 6 m. The zone is a 0.8 m square at the origin with back walls to its south and east; lookout 1 stands 2 m north of it and lookout 2 stands 2 m west. The two bars mark where a screen may stand, at $y = 1.15$ m and $x = -1.15$ m; at most one holds one. The solid route goes from the start to lookout 1, and the dashed one, from lookout 1 to lookout 2, is driven only behind a screen.

The back walls are not decoration: a laser certifies a cell free only along a ray that ends on something, and the rays that cross the zone end on them. In the blocked worlds a 0.35 m crate stands in the zone. The robot is a TurtleBot3 Burger mapping the hall with slam_toolbox from a start in the north-east corner, perturbed by the seed by up to 0.2 m on each axis and 10° in heading. Six worlds rotate over the seeds, seed s taking world $s \bmod 6$, so thirty seeds give each world five runs.

4 No conformant plan, and a policy

Proposition 1 (No conformant plan)**.** No sequence of the domain's actions is applicable in all six initial worlds and leaves Kw_{r1} *blocked-z* true in all of them.

Proof. Initially Kw_{r1} *blocked-z* is false, since $r1$ relates a world where *blocked-z* holds to one where it does not. The only actions whose events differ on *blocked-z* are mirar-1 and mirar-2; every other action leaves $r1$'s relation between two worlds that differ on *blocked-z* in place, so a sequence that reaches the goal contains a look. Consider the first look in it, mirar-k. For the sequence to be applicable in every world, its precondition must hold in the state reached from every world, and in particular from a world where *tapado-k* holds. No action changes *tapado-k*, so *tapado-k* still holds there, and since the frame is S5, $K_{r1}\neg$*tapado-k* implies $\neg$*tapado-k*: the precondition is false. Such a world exists for $k = 1$ and for $k = 2$. □

Proposition 2 (The policy)**.** The policy

$$\text{goto-m1};\ \text{vista-1};\ \begin{cases} \text{e-vista1-tapada}: & \text{goto-m2};\ \text{mirar-2} \\ \text{e-vista1-despejada}: & \text{mirar-1} \end{cases}$$

is applicable in all six worlds, reaches Kw_{r1} *blocked-z* in all six, and executes goto-m2 in exactly the two worlds where *tapado-1* holds.

Proof. goto-m1 and vista-1 are applicable everywhere. If *tapado-1* holds, e-vista1-tapada occurs and the designated worlds become the two where *tapado-1* holds; in both, $\neg$*tapado-2*, so $K_{r1}\neg$*tapado-2*

holds after goto-m2, which changes no screen, and mirar-2 is applicable. If ¬*tapado-1*, e-vista1-despejada occurs, the designated worlds are the four where ¬*tapado-1*, and mirar-1 is applicable. In either branch the look's two events split the designated worlds by *blocked-z*, and whichever occurs, $r1$ relates only worlds that agree on it. goto-m2 lies only on the first branch. □

The precondition $K_{r1}\neg$*tapado-k* is what makes the difference between the two propositions: it is the weakest condition under which a look from k is certain to decide the zone, and it is unknown until the robot has looked. The branch on vista-1 is where the robot finds out, and the elimination in the first branch, from *tapado-1* and $C\neg(\textit{tapado-1} \wedge \textit{tapado-2})$ to $K_{r1}\neg$*tapado-2*, is why no check of lookout 2 is needed.

Classical plans. A classical planner given the task without knowledge has looks with no modal precondition: `look_into` requires only that the robot stand at a lookout, and lasts its declared 8 s. The two plans compared are POPF's for the goal of having looked from lookout 1, a sequence of two actions, and for the goal of having looked from both, a sequence of four. In the world, a look from lookout k behind a screen observes no cell of the zone. So, as in the proof of Proposition 1, a sequence that leaves the zone observed in every world contains a look from lookout 2 and drives there in every world, and one that does not drive there leaves the zone unobserved in the two worlds with a screen at lookout 1.

5 On the robot

The harness launches the PlanSys2 stack with ePlanSys's epistemic state, perception and executor, slam_toolbox, and the action nodes of the six-room demonstration, reused with other parameters; it adds a recorder of the map, the actions and the true pose, and a mission node that sends the order and records the outcome. The perception watches three regions of the occupancy grid: the strip where screen 1 would stand, and the zone, once for each look. A region is blocked when one of its cells is occupied, clear when every cell is free, and undecided otherwise. Each region answers one sensing action, and the epistemic state applies an observation only when its event is applicable.

The epistemic look action turns the robot to the region and returns when the epistemic state reports Kw_{r1} of the action's atom. On the recorded runs that happened on the first grid after the robot faced the region, and the action took between 0.2 and 2.3 s. A run is *backed* when the map the robot built classifies the zone, by the same rule, as the robot reported.

6 Thirty seeds

Each seed fixes the world, the start pose and the laser noise, 0.01 m, and is run under all three formulations, so the comparison is paired. Times are in simulated seconds, from the order to the end of the mission.

	reports success	backed by the map	to lookout 2	median time	median path
epistemic	30/30	**30/30**	**10/30**	**15.1 s**	2.71 m
classical, one lookout	30/30	20/30	0/30	22.8 s	2.63 m
classical, both lookouts	30/30	30/30	30/30	51.2 s	6.28 m

Table 1: The batch, thirty seeds per formulation.

world	epistemic looks	epistemic	classical, one	classical, both
clear, screen 1	vista-1, mirar-2	5/5 · 34.9 s	**0/5** · 22.5 s	5/5 · 51.2 s
blocked, screen 1	vista-1, mirar-2	5/5 · 35.6 s	**0/5** · 22.8 s	5/5 · 51.9 s
clear, no screen	vista-1, mirar-1	5/5 · 14.5 s	5/5 · 22.6 s	5/5 · 50.8 s
blocked, no screen	vista-1, mirar-1	5/5 · 14.4 s	5/5 · 22.4 s	5/5 · 50.7 s
clear, screen 2	vista-1, mirar-1	5/5 · 15.1 s	5/5 · 23.0 s	5/5 · 51.4 s
blocked, screen 2	vista-1, mirar-1	5/5 · 14.9 s	5/5 · 22.8 s	5/5 · 51.3 s

Table 2: By world: runs backed by the map, and the median time.

The ten unbacked classical runs are the ten behind screen 1: the map holds the screen's place as blocked and no cell of the zone as observed, and the mission reports success. On backed success the discordant pairs are 10 to 0, and McNemar's exact test [5] gives $p = 0.002$. Against the classical plan through both lookouts every paired difference in time favours the policy, with a Hodges–Lehmann estimate [4] of 26.6 s; against the one-lookout plan the differences change sign with the world, the policy being faster in the four worlds it decides from lookout 1, by about 8 s, since its looks end on knowledge, and slower by about 12 s in the two it is forced to lookout 2. No run of any formulation reported the zone with the wrong class, and every mission ended in success.

7 The film

Four takes, recorded one at a time on a ROS domain and Gazebo port of their own, with the harness unchanged and headless: a Gazebo client and RViz on a virtual display, the world held paused until both had drawn, and the Gazebo camera cut on the executor's own actions.

take	world	looks	reports	sim time	the zone on the map
epistemic, seed 1	clear, screen 1	vista-1, mirar-2	SUCCESS	34.9 s	256 of 256 cells, clear
classical, one, seed 1	clear, screen 1	mirar-1	SUCCESS	22.3 s	0 of 256 cells, unknown
classical, both, seed 1	clear, screen 1	mirar-1, mirar-2	SUCCESS	51.1 s	256 of 256 cells, clear
epistemic, seed 3	clear, no screen	vista-1, mirar-1	SUCCESS	14.3 s	256 of 256 cells, clear

Table 3: The film's takes.

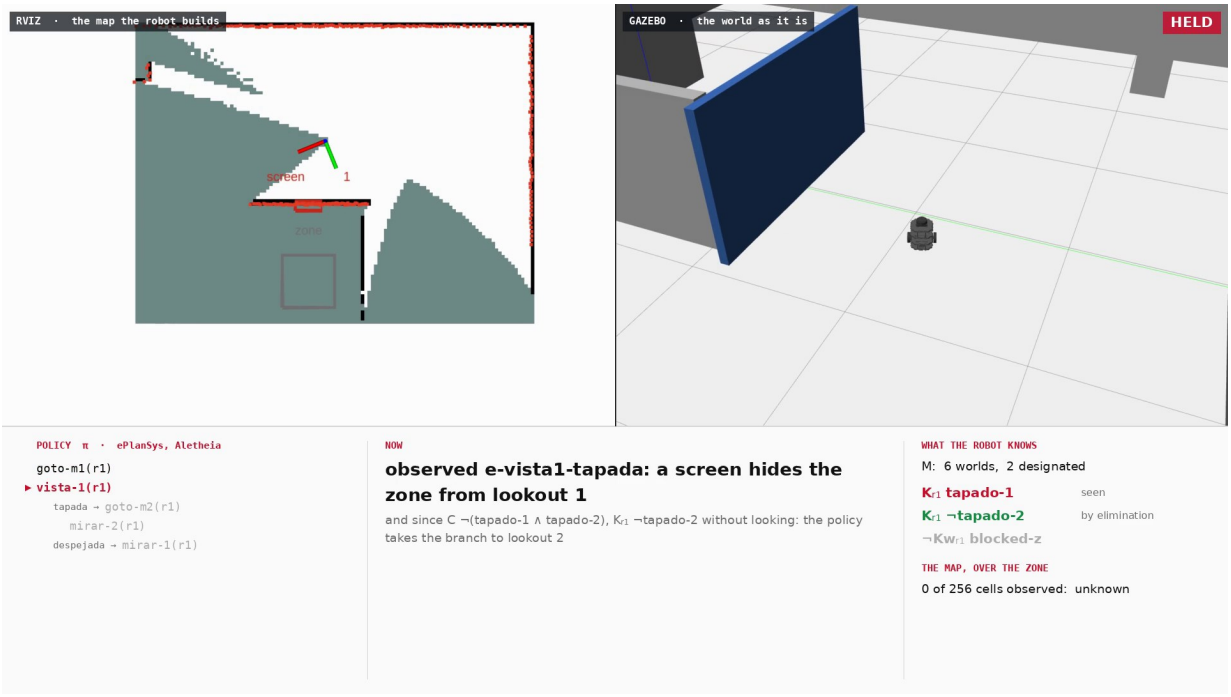


Figure 2: The film at the screen on seed 1, held. Left, RViz: the map, with the screen's place outlined red, blocked, and the zone grey, undecided. Right, Gazebo: the robot at lookout 1 before the screen. Under both: the policy with vista-1 running and its branches, the observation, and what the robot knows: K_{r1} *tapado-1* seen, $K_{r1}\neg$*tapado-2* by elimination, and the zone still unknown, with 0 of its 256 cells observed.

Every figure on the film's panels is a line one of the runs wrote. The model's counts on the recorded runs went from 6 designated worlds to 2 at the screen and to 1 at the zone on seed 1, and from 6 to 4 and to 2 on seed 3, where whether a screen stands at lookout 2 remains open and no longer matters.

Two timings found on the way. The executor reports an observed outcome after the epistemic state has applied it; on seed 3 after the zone's observation too, since there the zone was in view from lookout 1, and the state applied e-mirar1-libre at the same instant as e-vista1-despejada, as soon as $K_{r1}\neg$*tapado-1* made it applicable. mirar-1 then ran and returned in 0.2 s. A first cut of the film timed its captions from the executor's reports and so showed the model a step ahead at the clear view; the published film times them from the epistemic state. The log of every epistemic run also repeats that mirar-1_r1 is not applicable: it is the zone's region for the look not in progress, offered and refused, and it changes nothing.

8 What this does not show

- A stronger classical agent. A classical planner inside a loop that monitors execution and replans when a look does not decide would also reach the zone behind a screen. It would need the condition the epistemic domain writes as a precondition written into the monitor, and what a look decides written into the replanning.
- Noise that confuses a screen with a clear view. At 0.01 m the regions decided correctly in every run; noisier lasers were not run here.
- More than one robot, or more than two lookouts. Neither was built or measured.
- Localisation. The map is built with the simulator's odometry, and the world frame is the map frame.

References

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